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$$A = (a_{ij})_{n \times n}$$
$$\varphi(\lambda) = \det(\lambda I - A) = 0$$

(9.1.1)

$$\varphi(\lambda)$$
$$A$$
$$\varphi(\lambda) = \lambda^n + c_1 \lambda^{n-1} + \cdots + c_n = 0.$$

$$\varphi(\lambda)$$
$$n$$
$$A$$
$$\lambda$$
$$A$$
$$(\lambda I - A)x = 0$$

(9.1.2)

$$Ax = \lambda x$$
$$x$$
$$A$$
$$\lambda$$
$$\lambda_i \; (i = 1, 2, \cdots, n)$$
$$A$$
$$1^\circ \quad \sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \operatorname{tr} A;$$
$$2^\circ \quad \det A = \lambda_1 \lambda_2 \cdots \lambda_n.$$

$$A$$
$$B$$
$$T$$
$$B = T^{-1}AT$$

$$1^\circ$$

$$A$$
$$B$$

$$2^\circ$$

$$x$$

$$B$$

$$Tx$$

$$A$$

$$A = (a_{ij})_{n \times n}$$

A

$$|\lambda - a_{ii}| \leq \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \qquad (i = 1, 2, \cdots, n).$$

λ

A

x

$(\lambda I - A)x = 0,$

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$$x = (x_1, x_2, \cdots, x_n)^{\mathrm{T}} \neq 0$$

$$|x_i| = \max_k |x_k|$$

$$x_i \neq 0$$

i

$$(\lambda - a_{ii})x_i = \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij}x_j$$

$$|x_j/x_i| \leq 1 \ (j \neq i)$$

$$|\lambda - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|, \quad |x_j/x_i| \leq \sum_{j \neq i} |a_{ij}|.$$

λ

a_{ii}

$$\sum_{j \neq i} |a_{ij}|$$

A

A

i

i

A

n

x

$$R(x) = \frac{(Ax,x)}{(x,x)}$$

x

$$A \in \mathbf{R}^{n \times n}$$

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$$

$$x_1,x_2,\cdots,x_n$$

$$(x_i,x_j)=\delta_{ij}$$

$$1^\circ \quad \lambda_n \leq \frac{(Ax,x)}{(x,x)} \leq \lambda_1 \qquad (\text{对于任何非零} x \in \mathbf{R}^n) \ ;$$

$$2^{\circ} \quad \lambda_1 = \max_{\substack{x \in \mathbf{R}^n \\ x \neq 0}} \frac{(Ax,x)}{(x,x)};$$

$$3^{\circ} \quad \lambda_n = \min_{\substack{x \in \mathbf{R}^n \\ x \neq 0}} \frac{(Ax,x)}{(x,x)}.$$

$$1^{\circ}$$

$$2^{\circ}$$

$$3^{\circ}$$

$$x\neq 0$$

$$\mathbf{R}^n$$

$$x=\sum_{i=1}^na_ix_i,\quad \|x\|_2=\left(\sum_{i=1}^na_i^2\right)^2\neq 0,$$

$$\frac{(Ax,x)}{(x,x)}=\frac{\sum\limits_{i=1}^na_i^2\lambda_i}{\sum\limits_{i=1}^na_i^2},$$

$$1^{\circ}$$

$$1^{\circ}$$

$$\lambda_n$$

$$\lambda_1$$

$$A$$

$$n=2,3$$

$$\varphi(\lambda)=0$$

$$n$$

$$\varphi(\lambda)$$

$$\varphi(\lambda)$$

$$A$$

$$\sum_{j\neq i}|a_{ij}|$$

$$|x_j/x_i|\leq \sum_{j\neq i}|a_{ij}|$$

$$x_i$$

$$|\lambda-a_{ii}|\leq \sum_{j\neq i}|a_{ij}|\,|x_j/x_i|\leq \sum_{j\neq i}|a_{ij}|$$

$$2$$

$$\|x\|_2^2=\sum_i a_i^2$$

$$1/2$$

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$$A$$

$$A$$

$$A=(a_{ij})_n$$

$$\lambda_1,\lambda_2,\cdots,\lambda_n$$

$$x_1,x_2,\cdots,x_n$$

$$A$$

$$|\lambda_1|>|\lambda_2|\geq|\lambda_3|\geq\cdots\geq|\lambda_n|.\tag{9.2.1}$$

$$v_0$$

$$A$$

$$\left\{\begin{array}{l}v_1=Av_0,\\v_2=Av_1=A^2v_0,\\ \qquad\vdots\\v_{k+1}=Av_k=A^{k+1}v_0,\\ \qquad\vdots\end{array}\right.\tag{9.2.2}$$

$$v_0$$

$$v_0=a_1x_1+a_2x_2+\cdots+a_nx_n\quad(\text{设}a_1\neq 0)\quad,\tag{9.2.3}$$

$$v_k=Av_{k-1}=A^kv_0=a_1\lambda_1^kx_1+a_2\lambda_2^kx_2+\cdots+a_n\lambda_n^kx_n$$

$$=\lambda_1^k\left[a_1x_1+\sum_{i=2}^na_1(\lambda_i/\lambda_1)^kx_i\right]=\lambda_1^k(a_1x_1+\varepsilon_k),$$

$$\varepsilon_k=\sum_{i=2}^na_1(\lambda_i/\lambda_1)^kx_i$$

$$|\lambda_i/\lambda_1|<1\;(i=2,3,\cdots,n)$$

$$\varepsilon_k\rightarrow 0\;(k\rightarrow\infty)$$

$$\lim_{k\rightarrow\infty}\frac{v_k}{\lambda_1^k}=a_1x_1.\tag{9.2.4}$$

$$\frac{v_k}{\lambda_1^k}$$

$$A$$

$$\lambda_1$$

$$k$$

$$v_k\approx a_1\lambda_1^kx_1,\tag{9.2.5}$$

$$v_k$$

$$\lambda_1$$

$$\lambda_1$$

$$(v_k)_i$$

$$v_k$$

$$\frac{(v_{k+1})_i}{(v_k)_i}=\lambda_1\left\{\frac{a_1(x_1)_i+(\varepsilon_{k+1})_i}{a_1(x_1)_i+(\varepsilon_k)_i}\right\},\tag{9.2.6}$$
$$v_k$$
$$\varepsilon_k$$
$$a_1$$
$$a_i\lambda_i^kx_i$$
$$a_i$$

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$$\lim_{k\rightarrow\infty}\frac{(v_{k+1})_i}{(v_k)_i}=\lambda_1,\tag{9.2.7}$$
$$v_0$$
$$A$$
$$A^k$$
$$\{v_k\}$$
$$A$$
$$\lambda_1$$
$$(v_{k+1})_i/(v_k)_i\rightarrow\lambda_1$$
$$r=\lambda_2/\lambda_1$$
$$r$$
$$r=\lambda_2/\lambda_1\approx1$$
$$A\in\mathbf{R}^{n\times n}$$
$$n$$
$$\lambda_1$$
$$|\lambda_1|>|\lambda_2|\geq|\lambda_3|\geq\cdots\geq|\lambda_n|,$$
$$v_0\;(a_1\neq0)$$
$$A$$
$$\lambda_1=\lambda_2=\cdots=\lambda_r$$
$$|\lambda_r|>|\lambda_{r+1}|\geq\cdots\geq|\lambda_n|$$
$$A$$
$$n$$
$$\lambda_1$$
$$r$$
$$x_1,x_2,\cdots,x_r$$

$$\begin{aligned} v_k &= A^k v_0 = \lambda_1^k \left\{ \sum_{i=1}^r a_i x_i + \sum_{i=r+1}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right\}, \quad \lim_{k \rightarrow \infty} \frac{v_k}{\lambda_1^k} = \sum_{i=1}^r a_i x_i \quad \left(\text{设 } \sum_{i=1}^r a_i x_i \neq 0 \right). \\ A \\ A \\ \lambda_1 \\ |\lambda_1| > 1 \\ |\lambda_1| < 1 \\ v_k \\ k \rightarrow \infty \\ v \neq 0 \\ u &= \frac{v}{\max(v)} \\ \max(v) \\ v \\ v_0 \neq 0 \ (a_1 \neq 0) \\ \left\{ \begin{array}{ll} v_1 = Au_0 = Av_0, & u_1 = \frac{v_1}{\max(v_1)} = \frac{Av_0}{\max(Av_0)}, \\ v_2 = Au_1 = \frac{A^2 v_0}{\max(Av_0)}, & u_2 = \frac{v_2}{\max(v_2)} = \frac{A^2 v_0}{\max(A^2 v_0)}, \\ \vdots & \vdots \\ v_k = \frac{A^k v_0}{\max(A^{k-1} v_0)}, & u_k = \frac{A^k v_0}{\max(A^k v_0)}. \end{array} \right. \\ Av_0 = \sum_{i=1}^n a_i \lambda_i^k x_i = \lambda_i^k \left[a_1 x_1 + \sum_{i=2}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right], \\ Av_0 \\ \lambda_i^k \\ v_0 = \sum_i a_i x_i \\ Ax_i = \lambda_i x_i \\ A^k v_0 \\ \lambda_1^k \end{aligned}$$

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$$\begin{aligned} u_k &= \frac{A^k v_0}{\max(A^k v_0)} = \frac{\lambda_i^k \left[a_1 x_1 + \sum_{i=2}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right]}{\max \left[\lambda_1^k \left(a_1 x_1 + \sum_{i=2}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right) \right]} \\ &= \frac{\left[a_1 x_1 + \sum_{i=2}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right]}{\max \left[a_1 x_1 + \sum_{i=2}^n a_i \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right]} \rightarrow \frac{x_1}{\max(x_1)} \quad (k \rightarrow \infty). \end{aligned}$$

(9.2.8)

$$v_k=\frac{\lambda_1^k\left[a_1x_1+\sum_{i=2}^na_i\left(\frac{\lambda_i}{\lambda_1}\right)^kx_i\right]}{\max\left[\lambda_1^{k-1}a_1x_1+\sum_{i=2}^na_i\left(\frac{\lambda_i}{\lambda_1}\right)^{k-1}x_i\right]},$$

$$\max(v_k)=\frac{\lambda_1\max\left[a_1x_1+\sum_{i=2}^na_i\left(\frac{\lambda_i}{\lambda_1}\right)^kx_i\right]}{\max\left[a_1x_1+\sum_{i=2}^na_i\left(\frac{\lambda_i}{\lambda_1}\right)^{k-1}x_i\right]}\rightarrow\lambda_1\quad(k\rightarrow\infty).$$

$$r=\lambda_2/\lambda_1$$

$$A\in\mathbf{R}^{n\times n}$$

$$n$$

$$\lambda_1$$

$$|\lambda_1|>|\lambda_2|\geq|\lambda_3|\geq\cdots\geq|\lambda_n|$$

$$v_0=u_0~(a_1\neq 0)$$

$$\begin{cases} v_0=u_0\neq 0,\\ v_k=Au_{k-1},\qquad (k=1,2,\cdots),\\ u_k=\dfrac{v_k}{\max(v_k)} \end{cases}\tag{9.2.9}$$

$$\lim_{k\rightarrow\infty}u_k=\frac{x_1}{\max(x_1)},\quad\lim_{k\rightarrow\infty}\max(v_k)=\lambda_1.$$

$$A=\begin{pmatrix}1.0&1.0&0.5\\1.0&1.0&0.25\\0.5&0.25&2.0\end{pmatrix}$$

$$u_k$$

$$\lambda_1\approx 2.536\,532\,3$$

$$(0.748\,2,\;0.649\,7,\;1)^{\mathrm{T}}$$

$$\lambda_1$$

$$\lambda_1=2.536\,525\,8,\quad \widetilde{x}_1=(0.748\,221\,16,\;0.649\,661\,16,\;1)^{\mathrm{T}}.$$

$$u_k$$

$$\lambda_i^k$$

$$\lambda_1^k$$

$$A^kv_0$$

$$\lambda_1^k$$

$$v_k$$

$$\lambda_1^{k-1}$$

$$a_1x_1$$

$$v_k=A^kv_0/\max(A^{k-1}v_0)$$

$$\lambda_1^{k-1}$$

$$a_1x_1+\sum_{i=2}^na_i(\lambda_i/\lambda_1)^{k-1}x_i$$

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$$k$$

$$u_k^{\mathrm{T}}$$

$$\max(v_k)$$

$$(1,1,1)$$

$$(0.909\,1,\;0.818\,2,\;1)$$

$$2.750\,000\,0$$

$$(0.765\,1,\;0.667\,4,\;1)$$

$$2.558\,791\,8$$

$$(0.749\,4,\;0.650\,8,\;1)$$

$$2.538\,002\,9$$

$$(0.748\,3,\;0.649\,7,\;1)$$

$$2.536\,625\,6$$

$$(0.748\,3,\;0.649\,7,\;1)$$

$$2.536\,584\,0$$

$$(0.748\,2,\;0.649\,7,\;1)$$

$$2.536\,559\,8$$

$$(0.748\,2,\;0.649\,7,\;1)$$

$$2.536\,545\,6$$

$$(0.748\,2,\;0.649\,7,\;1)$$

$$2.536\,537\,4$$

$$(0.748\,2,\;0.649\,7,\;1)$$

$$2.536\,532\,3$$

$$A$$

$$r=\frac{\lambda_1}{\lambda_2}$$

$$r$$

$$B=A-pI$$

$$p$$

$$A$$

$$\lambda_1,\lambda_2,\cdots,\lambda_n$$

$$B$$

$$\lambda_1-p,\lambda_2-p,\cdots,\lambda_n-p$$

$$A,B$$

$$A$$

$$\lambda_1$$

$$p$$

$$\lambda_1-p$$

$$B$$

$$\left|\frac{\lambda_2-p}{\lambda_1-p}\right|<\left|\frac{\lambda_2}{\lambda_1}\right|.$$

$$B$$

$$B$$

$$\lambda_1-p$$

$$A$$

$$A=(a_{ij})_4$$

$$\lambda_j=15-j\quad (j=1,2,3,4)$$

$$r=\lambda_2/\lambda_1\approx 0.9$$

$$B=A-pI\quad (p=12),$$

$$B$$

$$\mu_1=2,\; \mu_2=1,\; \mu_3=0,\; \mu_4=-1$$

$$B$$

$$\mu_1$$

$$\left|\frac{\mu_2}{\mu_1}\right|=\left|\frac{\lambda_2-p}{\lambda_1-p}\right|=\frac{1}{2}<\left|\frac{\lambda_2}{\lambda_1}\right|\approx 0.9.$$

$$r=\lambda_1/\lambda_2$$

$$r=\lambda_2/\lambda_1$$

$$(\lambda_i/\lambda_1)^k$$

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$$p$$

$$p$$

$$A$$

$$p$$

$$\lambda_1$$

$$A$$

$$\lambda_1>\lambda_2\geq\cdots\geq\lambda_{n-1}>\lambda_n,\tag{9.2.10}$$

$$p$$

$$B=A-pI$$

$$\lambda_1-p$$

$$\lambda_n-p$$

$$\lambda_1$$

$$x_1$$

$$p$$

$$\begin{aligned} &|\lambda_1-p|>|\lambda_n-p|\\ \omega &= \max\left\{\frac{|\lambda_2-p|}{|\lambda_1-p|},\frac{|\lambda_n-p|}{|\lambda_1-p|}\right\}=\min,\\ &\lambda_2-p=-(\lambda_n-p)\\ p &= \frac{\lambda_2+\lambda_n}{2}\equiv p^* \end{aligned}$$

$$\begin{aligned} &\omega\\ \frac{\lambda_2-p^*}{\lambda_1-p^*} &= -\frac{\lambda_n-p^*}{\lambda_1-p^*}\equiv \frac{\lambda_2-\lambda_n}{2\lambda_1-\lambda_2-\lambda_n}.\\ &A \end{aligned}$$

$$\lambda_2,\lambda_n$$

$$p^*$$

$$\begin{aligned} &\lambda_n\\ p &= \frac{\lambda_1+\lambda_{n-1}}{2}=p^*\\ &\lambda_n \end{aligned}$$

$$A$$

$$B=A-pI$$

$$\begin{aligned} &p=0.75\\ B &= \begin{pmatrix} 0.25 & 1 & 0.5 \\ 1 & 0.25 & 0.25 \\ 0.5 & 0.25 & 1.25 \end{pmatrix}.\\ &B \end{aligned}$$

$$k$$

$$u_k^{\mathrm{T}}$$

$$\max(v_k)$$

$$(1,1,1)$$

$$(0.751\,6,\;0.652\,2,\;1)$$

$$1.791\,401\,1$$

$$(0.749\,1,\;0.651\,1,\;1)$$

$$1.788\,844\,3$$

$$(0.748\,8,\;0.650\,1,\;1)$$

$$1.787\,330\,0$$

$$(0.748\,4,\;0.649\,9,\;1)$$

$$1.786\,915\,2$$

$$(0.748\,3,\;0.649\,7,\;1)$$

$$\begin{array}{c} 1.786\,658\,7 \\ (0.748\,2,\; 0.649\,7,\; 1) \\ 1.786\,591\,4 \\ B \\ \mu_1 \approx 1.786\,591\,4 \\ A \\ \lambda_1 \\ \lambda_1 \approx \mu_1 + 0.75 = 2.536\,591\,4. \\ \mu_1 = 1.786\,525\,8 \\ \lambda_1 = 2.536\,525\,8 \end{array}$$

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$$\begin{array}{c} A \\ p \\ A \\ A \\ \lambda_1 \\ \lambda_n \\ A \\ A \in \mathbf{R}^{n \times n} \\ |\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n| \\ (x_i, x_j) = \delta_{ij} \\ A \\ \lambda_1 \\ u_k \\ \lambda_1 \\ \frac{(Au_k, u_k)}{(u_k, u_k)} = \lambda_1 + O\left(\left(\frac{\lambda_2}{\lambda_1}\right)^{2k}\right). \\ u_k = \frac{A^k u_0}{\max(A^k u_0)} \\ v_{k+1} = Au_k = \frac{A^{k+1} u_0}{\max(A^k u_0)} \\ \frac{(Au_k, u_k)}{(u_k, u_k)} = \frac{(A^{k+1} u_0, A^k u_0)}{(A^k u_0, A^k u_0)} = \frac{\sum_{j=1}^n a_j^2 \lambda_j^{2k+1}}{\sum_{j=1}^n a_j^2 \lambda_j^{2k}} = \lambda_1 + O\left(\left(\frac{\lambda_2}{\lambda_1}\right)^{2k}\right). \end{array} \tag{9.2.11}$$

$$A\in\mathbf{R}^{n\times n}$$

$$A$$

$$|\lambda_1|\geq |\lambda_2|\geq \cdots \geq |\lambda_n|$$

$$x_1,x_2,\cdots,x_n$$

$$A^{-1}$$

$$\left|\frac{1}{\lambda_n}\right|\geq \left|\frac{1}{\lambda_{n-1}}\right|\geq \cdots \geq \left|\frac{1}{\lambda_1}\right|$$

$$x_n,x_{n-1},\cdots,x_1$$

$$A$$

$$\lambda_n$$

$$A^{-1}$$

$$A^{-1}$$

$$A^{-1}$$

$$1/\lambda_n$$

$$A$$

$$\lambda_n$$

$$v_0=u_0\neq 0$$

$$\begin{cases} v_k=A^{-1}u_{k-1},\\ u_k=\frac{v_k}{\max(v_k)} \end{cases} \quad (k=1,2,\cdots).$$

$$v_k$$

$$Av_k=u_{k-1}$$

$$1^\circ$$

$$A$$

$$n$$

$$2^\circ$$

$$A$$

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$$|\lambda_1|\geq |\lambda_2|\geq \cdots \geq |\lambda_{n-1}|>|\lambda_n|>0,$$

$$u_0=v_0\;(a_n\neq 0)$$

$$\{v_k\},\{u_k\}$$

$$\lim_{k\rightarrow\infty}u_k=\frac{x_n}{\max(x_n)},\quad\lim_{k\rightarrow\infty}\max(v_k)=\frac{1}{\lambda_n}.$$

$$\left|\frac{\lambda_n}{\lambda_{n-1}}\right|$$

$$\left(A-pI\right)^{-1}$$

$$\frac{1}{\lambda_1-p},\frac{1}{\lambda_2-p},\cdots,\frac{1}{\lambda_n-p}$$
$$x_1,x_2,\cdots,x_n$$
$$(A-pI)^{-1}$$
$$\left\{\begin{array}{l} u_0=v_0\neq 0\quad(\text{初始向量})\;,\\ v_k=(A-pI)^{-1}u_{k-1},\qquad\qquad(k=1,2,\cdots).\\ u_k=\frac{v_k}{\max(v_k)}\end{array}\right.$$

(9.2.12)

$$p$$
$$A$$
$$\lambda_j$$
$$|\lambda_j-p|<|\lambda_i-p|\;(i\neq j)$$
$$\frac{1}{\lambda_j-p}$$
$$(A-pI)^{-1}$$
$$A\in\mathbf{R}^{n\times n}$$
$$n$$
$$x_1,x_2,\cdots,x_n$$
$$u_0=\sum_{i=1}^na_ix_i\quad(a_i\neq 0),\quad v_k=\frac{(A-pI)^{-k}u_0}{\max((A-pI)^{-(k-1)}u_0)},$$
$$u_k=\frac{(A-pI)^{-k}u_0}{\max((A-pI)^{-k}u_0)},$$
$$(A-pI)^{-k}u_0=\sum_{i=1}^na_i(\lambda_i-p)^{-k}x_i.$$
$$1^\circ$$
$$A\in\mathbf{R}^{n\times n}$$
$$n$$
$$A$$
$$\lambda_i$$
$$x_i\;(i=1,2,\cdots,n)$$
$$2^\circ$$
$$p$$
$$\lambda_j$$
$$(A-pI)^{-1}$$
$$|\lambda_j-p|<|\lambda_i-p|\;(i\neq j)$$
$$3^\circ$$
$$u_0=\sum_{i=1}^na_ix_i\neq 0$$
$$(a_i\neq 0)$$
$$\{v_k\},\{u_k\}$$

$$\lim_{k\rightarrow\infty}u_k=\frac{x_j}{\max(x_j)},$$
$$\lim_{k\rightarrow\infty}\max(v_k)=\frac{1}{\lambda_j-p},\quad\text{即}\quad p+\frac{1}{\max(v_k)}\rightarrow\lambda_j\quad(\text{当}k\rightarrow\infty)\quad,$$
$$r=\max_{i\neq j}\left|\frac{\lambda_j-p}{\lambda_i-p}\right|$$

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$$A-pI$$

$$p\approx\lambda_j$$

$$x_j$$

$$p$$

$$\lambda_j$$

$$r$$

$$v_k$$

$$(A-pI)v_k=u_{k-1}$$

$$(A-pI)$$

$$P(A-pI)=LU,$$

$$P$$

$$v_k$$

$$Ly_k=Pu_{k-1},Uv_k=y_k$$

$$\left\{\begin{array}{l} Ly_k=Pu_{k-1},\\ Uv_k=y_k,\\ u_k=\frac{v_k}{\max(v_k)} \end{array}\right.\qquad (k=1,2,\cdots).$$

(9.2.13)

$$v_0=u_0$$

$$u_0$$

$$Uv_1=L^{-1}Pu_0=(1,1,\cdots,1)^{\rm T},$$

(9.2.14)

$$v_1$$

$$A=\begin{pmatrix}2&1&0\\1&3&1\\0&1&4\end{pmatrix}$$

$$\lambda=1.267\,9$$

$$\lambda_3=3-\sqrt{3}$$

$$A-pI$$

$$p=1.267\,9$$

$$P(A-pI)=LU,$$

$$P=\begin{pmatrix}0&1&0\\0&0&1\\1&0&0\end{pmatrix},$$

$$L=\begin{pmatrix}1&0&0\\0&1&0\\0.732\,1&-0.268\,07&1\end{pmatrix},\quad U=\begin{pmatrix}1&1.732\,1&1\\0&1&2.732\,1\\0&0&0.294\,05\times10^{-3}\end{pmatrix}.$$

$$Uv_1=(1,1,1)^{\mathrm{T}}$$

$$v_1=(12\,692,\,-9\,290.3,\,3\,400.8)^{\mathrm{T}},\quad u_1=(1,\,-0.731\,98,\,0.267\,95)^{\mathrm{T}},$$

$$LUv_2=Pu_1$$

$$v_2=(20\,404,\,-14\,937,\,5\,467.4)^{\mathrm{T}},\quad u_2=(1,\,-0.732\,06,\,0.267\,96)^{\mathrm{T}}.$$

$$\lambda_3$$

$$x_3=(1,\,1-\sqrt{3},\,2-\sqrt{3})^{\mathrm{T}}\approx(1,\,-0.732\,05,\,0.267\,95)^{\mathrm{T}}.$$

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$$u_2$$

$$x_3$$

$$A\in\mathbf{R}^{n\times n}$$

$$A\in\mathbf{R}^{n\times n}$$

$$R$$

$$R^{\mathrm{T}}AR=\begin{pmatrix}T_{11}&T_{12}&\cdots&T_{1s}\\&T_{22}&\cdots&T_{2s}\\&&\ddots&\vdots\\&&&T_{ss}\end{pmatrix},$$

$$A$$

$$A$$

$$B$$

$$i>j+1$$

$$b_{ij}=0$$

$$B$$

$$B=\begin{pmatrix}b_{11}&b_{12}&\cdots&b_{1n}\\b_{21}&b_{22}&\cdots&b_{2n}\\&\ddots&\ddots&\vdots\\&&b_{n,n-1}&b_{nn}\end{pmatrix}.$$

$$w$$

$$\|w\|_2=1$$

$$H=I-2ww^{\mathrm{T}}$$

$$H(w)$$

$$H(w)=\begin{pmatrix}1-2w_1^2&-2w_1w_2&\cdots&-2w_1w_n\\-2w_2w_1&1-2w_2^2&\ddots&\vdots\\\vdots&\ddots&\ddots&-2w_{n-1}w_n\\-2w_nw_1&\cdots&-2w_nw_{n-1}&1-2w_n^2\end{pmatrix},$$

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$$w=(w_1,w_2,\cdots,w_n)^{\mathrm{T}}.$$

$$H$$

$$H^{\mathrm{T}}=H$$

$$H^{\mathrm{T}}H=I$$

$$H^2=I$$

$$H$$

$$H^{\mathrm{T}}H=H^2=(I-2ww^{\mathrm{T}})(I-2ww^{\mathrm{T}})=I-4ww^{\mathrm{T}}+4w(w^{\mathrm{T}}w)w^{\mathrm{T}}=I.$$

$$u\neq 0$$

$$H=I-2\frac{uu^{\mathrm{T}}}{\|u\|_2^2}$$

$$w$$

$$O$$

$$S:w^{\mathrm{T}}x=0.$$

$$v\in\mathbf{R}^n$$

$$v=x+y$$

$$x\in S,y\in S^\perp$$

$$Hx=(I-2ww^{\mathrm{T}})x=x-2ww^{\mathrm{T}}x=x.$$

$$y\in S^\perp$$

$$Hy=-y$$

$$v\in\mathbf{R}^n$$

$$Hv=x-y=v',$$

$$v'$$

$$v$$

$$S$$

$$w,v,x,y,O,S,v'$$

$$S$$

$$O$$

$$w$$

$$v=x+y$$

$$x$$

$$y$$

$$v'$$

$$x-y$$

$$O$$

$$v'$$

$$a\neq 0$$

$$H$$

$$H_a=\sigma e_1$$

$$x,y$$

$$n$$

$$\|x\|_2=\|y\|_2$$

$$H$$

$$Hx=y$$

$$w=\frac{x-y}{\|x-y\|_2}$$

$$H=I-2ww^{\mathrm{T}}=I-2\frac{(x-y)}{\|x-y\|_2^2}(x^{\mathrm{T}}-y^{\mathrm{T}}),$$

$$Hx=x-2\frac{x-y}{\|x-y\|_2^2}(x^{\mathrm{T}}-y^{\mathrm{T}})x=x-2\frac{(x-y)(x^{\mathrm{T}}x-y^{\mathrm{T}}x)}{\|x-y\|_2^2},$$

$$\|x-y\|_2^2=(x-y)^{\mathrm{T}}(x-y)=2(x^{\mathrm{T}}x-y^{\mathrm{T}}x),$$

$$Hx=x-(x-y)=y.$$

$$w=\frac{x-y}{\|x-y\|_2}$$

$$Hx=y$$

$$x\in\mathbf{R}^n\;(x\neq 0)$$

$$\sigma=\pm\|x\|_2$$

$$x\neq-\sigma e_1$$

$$H=I-2\frac{uu^{\mathrm{T}}}{\|u\|_2^2}\equiv I-\rho^{-1}uu^{\mathrm{T}},$$

$$Hx=-\sigma e_1$$

$$u=x+\sigma e_1$$

$$\rho=\|u\|_2^2/2$$

$$x=(\alpha_1,\alpha_2,\cdots,\alpha_n)^{\mathrm{T}}\neq 0,\quad u=(u_1,u_2,\cdots,u_n)^{\mathrm{T}},$$

$$H$$

$$a$$

$$H$$

$$H_a=\sigma e_1$$

$$Ha=\sigma e_1$$

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$$u = (\alpha_1 + \sigma, \alpha_2, \cdots, \alpha_n)^{\mathrm{T}},$$

$$\rho = \frac{1}{2}\|u\|_2^2 = \frac{1}{2}\left[(\alpha_1 + \sigma)^2 + \alpha_2^2 + \cdots + \alpha_n^2\right] = \sigma(\sigma + \alpha_1).$$

$$\sigma$$

$$\alpha_1$$

$$\alpha_1 + \sigma$$

$$\sigma$$

$$\alpha_1$$

$$\sigma = \operatorname{sgn}(\alpha_1)\|x\|_2.$$

$$x = (\alpha_1, \alpha_2, \cdots, \alpha_n)^{\mathrm{T}} \neq 0$$

$$\sigma, \rho$$

$$u$$

$$(I - \rho^{-1}uu^{\mathrm{T}})x = -\sigma e_1$$

$$u$$

$$x$$

$$\sigma = \operatorname{sgn}(\alpha_1)\left(\sum_{i=1}^n\alpha_i^2\right)^{\frac{1}{2}}$$

$$\alpha_1 \rightarrow u_1 = \alpha_1 + \sigma$$

$$\rho = \sigma u_1$$

$$\sigma$$

$$x$$

$$\eta = \max_i |\alpha_i|, \quad x' = \frac{x}{\eta},$$

$$\sigma' = \sigma/\eta, \quad H' = H.$$

$$x = (\alpha_1, \alpha_2, \cdots, \alpha_n)^{\mathrm{T}} \neq 0$$

$$H$$

$$\sigma$$

$$Hx = -\sigma e_1$$

$$u$$

$$x$$

$$\eta = \max_i |\alpha_i|$$

$$\alpha_i \leftarrow u_i = \frac{\alpha_i}{\eta} \quad (i = 1, 2, \cdots, n)$$

$$\sigma = \operatorname{sgn}(u_1)\left(\sum_{i=1}^nu_i^2\right)^{\frac{1}{2}}$$

$$u_1 \leftarrow u_1 + \sigma$$

$$\rho = \sigma u_1$$

$$\sigma \leftarrow \eta \sigma$$

$$HA$$

$$A=(a_1,a_2,\cdots,a_n)$$

$$a_i$$

$$A$$

$$i$$

$$HA=(Ha_1,Ha_2,\cdots,Ha_n),$$

$$HA$$

$$Ha_i=(I-\rho^{-1}uu^{\mathrm{T}})a_i=a_i-(\rho^{-1}u^{\mathrm{T}}a_i)u\quad(i=1,2,\cdots,n).$$

$$Ha_i$$

$$HA$$

$$2n^2$$

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$$A=\left(\begin{array}{c|ccc}a_{11}&a_{12}&\cdots&a_{1n}\\\hline a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&&\vdots\\a_{n1}&a_{n2}&\cdots&a_{nn}\end{array}\right)\equiv\left(\begin{array}{cc}a_{11}&A_{12}^{(1)}\\a_{21}^{(1)}&A_{22}^{(1)}\end{array}\right),$$

$$a_{21}^{(1)}\neq 0$$

$$R_1$$

$$R_1a_{21}^{(1)}=-\sigma_1e_1$$

$$\left\{\begin{array}{l} \sigma_1=\mathrm{sgn}(a_{21})\left(\sum_{i=2}^na_{i1}^2\right)^{\frac{1}{2}},\\ u_1=a_{21}^{(1)}+\sigma_1e_1,\\ \rho_1=\frac{1}{2}\|u_1\|_2^2=\sigma_1(\sigma_1+a_{21}),\\ R_1=I-\rho_1^{-1}u_1u_1^{\mathrm{T}}. \end{array}\right.\tag{9.3.1}$$

$$U_1=\begin{pmatrix}I&0\\0&R_1\end{pmatrix}$$

$$A_2=U_1A_1U_1=\begin{pmatrix}a_{11}&A_{21}^{(1)}R_1\\R_1a_{21}^{(1)}&R_1A_{22}^{(1)}R_1\end{pmatrix}\equiv\begin{pmatrix}A_{11}^{(2)}&a_{12}^{(2)}&A_{13}^{(2)}\\O&a_{22}^{(2)}&A_{23}^{(2)}\end{pmatrix},$$

$$A_{11}^{(2)}\in\mathbf{R}^{2\times1},\quad a_{22}^{(2)}\in\mathbf{R}^{n-2},\quad A_{23}^{(2)}\in\mathbf{R}^{(n-2)\times(n-2)}$$

$$k$$

$$A$$

$$k-1$$

$$A_k=U_{k-1}A_{k-1}U_{k-1}$$
$$=\left(\begin{array}{ccc|ccc}a_{11}&a_{12}^{(2)}&\cdots&a_{1k}^{(k)}&a_{1,k+1}^{(k)}&\cdots&a_{1n}^{(k)}\\-\sigma_1&a_{22}^{(2)}&\cdots&a_{2k}^{(k)}&a_{2,k+1}^{(k)}&\cdots&a_{2n}^{(k)}\\&&\ddots&\vdots&\vdots&&\vdots\\&&&-\sigma_{k-1}&a_{kk}^{(k)}&a_{k,k+1}^{(k)}&\cdots&a_{k,n}^{(k)}\\\hline&&&&a_{k+1,k}^{(k)}&a_{k+1,k+1}^{(k)}&\cdots&a_{k+1,n}^{(k)}\\&&&&\vdots&\vdots&&\vdots\\&&&&a_{nk}^{(k)}&a_{n,k+1}^{(k)}&\cdots&a_{nn}^{(k)}\end{array}\right)$$
$$\equiv\begin{pmatrix}A_{11}^{(k)}&a_{12}^{(k)}&A_{13}^{(k)}\\O&a_{22}^{(k)}&A_{23}^{(k)}\end{pmatrix},$$
$$A_{11}^{(k)}\in\mathbf{R}^{k\times(k-1)},\quad a_{22}^{(k)}\in\mathbf{R}^{n-k},\quad A_{23}^{(k)}\in\mathbf{R}^{(n-k)\times(n-k)}$$

$$a_{22}^{(k)}\neq 0$$

$$R_k$$

$$R_k a_{22}^{(k)}=-\sigma_k e_1$$
$$\left\{\begin{array}{l}\sigma_k=\text{sgn}(a_{k+1,k}^{(k)})\left(\sum_{i=k+1}^na_{ik}^2\right)^{\frac{1}{2}},\\u_k=a_{22}^{(k)}+\sigma_k e_1,\\\rho_k=\frac{1}{2}\|u_k\|_2^2=\sigma_k(\sigma_k+a_{k+1,n}^{(k)}),\\R_k=I-\rho_k^{-1}u_ku_k^{\text{T}}.\end{array}\right.$$

(9.3.2)

$$A_2$$

$$A_{21}^{(1)}R_1$$

$$A_{12}^{(1)}$$

$$U_1$$

$$A_{12}^{(1)}R_1$$

$$\sigma_k$$

$$a_{ik}^2$$

$$\rho_k$$

$$a_{k+1,n}^{(k)}$$

$$a_{22}^{(k)}=(a_{k+1,k}^{(k)},\ldots,a_{nk}^{(k)})^{\text{T}}$$

$$u_k=a_{22}^{(k)}+\sigma_k e_1$$

$$(a_{ik}^{(k)})^2$$

$$\rho_k=\sigma_k(\sigma_k+a_{k+1,k}^{(k)})$$

$$k$$

$$n$$